The Media Connectivity Association



OVERVIEW AND UPDATES · AI IN MEDIA SERVICES

AI in Media Services

Evaluation framework for AI/ML models in media processing for 3GPP systems

3GPP

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About the project

The reports the evaluation framework is built around

3GPP TR 26.847

Technical Report, V19.0.0

Evaluation of AI and ML in 5G media services: the evaluation testbed architectures and anchors, the metrics and the evaluation scenarios.

3GPP TR 26.927

Technical Report, V19.0.0

Study on AI and ML in 5G media services: the media use cases and the architectures for AI/ML model distribution and split operation.

3GPP TR 22.874

Technical Report, V18.2.0

Study on traffic characteristics and performance requirements for AI/ML model transfer in 5GS.

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The evaluation framework

What TR 26.847 defines for the evaluation of AI/ML in media services

Testbed architectures clause 4, common to every scenario unless stated

- Anchors: inference on the device, or in the network
- Split inference intermediate data testbed
- Model data testbed

Metrics clause 4.4, with the AI/ML task metrics

- AI/ML task metrics
- Video quality
- Complexity
- Bitrate
- Split model size
- Intermediate data size or bitrate
- Compressed model size
- Compressed intermediate data ratio
- Latency

Scenario template annex A, one entry per evaluation scenario

- Motivation and use case relevance
- Description
- Anchor model
- Testbed architecture and anchors
- Test configuration
- Metrics and requirements
- Test dataset and scripts
- Test conditions
- Interoperability

The functional side of AI/ML in media services is documented in TR 26.927; TR 26.847 is its evaluation companion.

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Scenarios

The four evaluation scenarios of TR 26.847 and the scenarios of the repository

In TR 26.847 the four evaluation scenarios of clause 5

- Compressed AI/ML model data for speech recognition
- Split inferencing for object detection and labelling
- Bit-incremental transmission and deployment of AI/ML models
- Real-time translation over IMS

Scripts in the repository read from the README and the scripts folders

- Compressed model transmission for speech recognition
- Split inferencing for object detection
- IMS real-time translation

Planned described in the README, with no scripts yet

- Split inferencing for human pose estimation
- Split inferencing for hand gesture recognition
- Video super resolution in video call
- Federated learning

The repository is an evaluation tool under development, not a service or a network function.

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What the scripts do

The three scenarios that have scripts today

COMPRESSED MODEL TRANSFER

Speech recognition models

A pre-trained model (wav2vec 2.0 or HuBERT) is compressed before it is sent to a device. A codec pipeline encodes and decodes it and reports its size and its recognition performance before and after.

SPLIT INFERENCING

Object detection

A pre-trained model detects objects in a video sequence and is split in two parts, one on the device and one in the network. Scripts convert and split a RetinaNet model and run it whole or split.

TRANSLATION

Translation models

Scripts train a Transformer translation model on Tatoeba sentence pairs and translate sentences, with split translation and a WebRTC translator. Four trained models are stored with Git LFS.

A Dockerfile builds the evaluation environment (Ubuntu 22.04, with Nvidia GPU support). The datasets are hosted outside the repository.

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How it grew

What each 3GPP release added to AI and ML in media services

Release 18

The SA1 study of the traffic of AI/ML model transfer, TR 22.874, is approved at version 18.0.0. Its Stage 1 requirements are in TS 22.261 (clause 6.40).

Release 19

The SA4 study FS_AI4Media: TR 26.927 and TR 26.847 are approved at SA#108 and created as version 19.0.0 (June 2025).

Release 20

Work plan of 28 September 2026. TR 26.927 recommends stage 3 work on AI/ML in IMS (TS 26.114 and TS 26.264) and TR 26.847 recommends a new evaluation study. These are recommendations, not decisions.

From the Standards pages of the website, and clause 7 of TR 26.927 and clause 6 of TR 26.847.

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Repositories

The repository of the AI/ML evaluation framework

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE
rt-ai-ml-evaluation-framework	the implementation of the AI/ML evaluation framework defined in 3GPP SA4 TR 26.847 (Release 19)	Compressed model transfer Split inferencing Translation scenario	On request Early Access

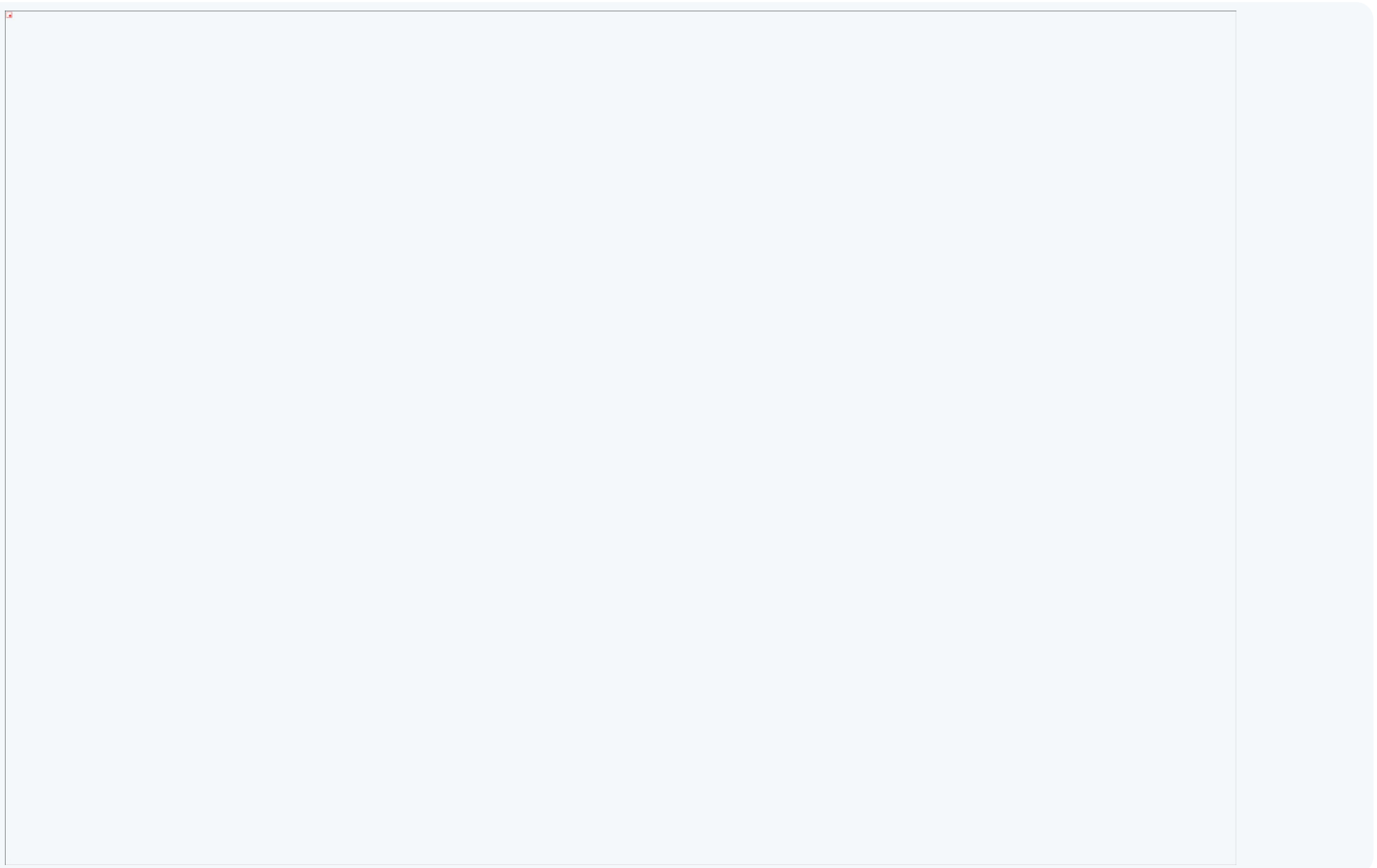
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● released ● no release yet

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1 **rt-ai-ml-evaluation-framework**

Boxes without a badge are not mapped to a repository.

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